THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Substances that taste sour are generally: 1

(a) salty **(b)** basic **(c)** acidic **(d)** neutral

2. Which of these is a laboratory acid? 1

(a) sulphuric acid **(b)** lactic acid **(c)** formic acid **(d)** citric acid

3. Red litmus turns blue in: 1

(a) salt solution **(b)** base **(c)** water **(d)** acid

4. Substances that feel soapy and taste bitter are generally: 1

(a) basic **(b)** acidic **(c)** sour **(d)** neutral

5. The formula of sulphuric acid is: 1

(a) H_2SO_4 **(b)** HNO_3 **(c)** $NaOH$ **(d)** HCl

6. Neutral substances: 1

(a) turn blue litmus red **(b)** turn red litmus blue

(c) do not change the colour of litmus **(d)** turn both red

7. Which of these is acidic? 1

(a) lime water **(b)** soap solution

(c) baking soda solution **(d)** lemon juice

8. The formula of sodium hydroxide is: 1

(a) $Ca(OH)_2$ **(b)** HCl **(c)** KOH **(d)** $NaOH$

9. Turmeric indicator turns red in: 1

(a) neutral **(b)** base **(c)** vinegar **(d)** acid

10. Which of these is basic? 1

(a) curd **(b)** vinegar

(c) baking soda solution **(d)** lemon juice

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What are indicators? Give two natural indicators. 2

12. Name two common bases used at home. 2

13. Write the formulae of hydrochloric acid and sodium hydroxide. 2

14. What is the effect of acids and bases on litmus paper? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how you will test whether a solution is acidic, basic or neutral using litmus paper. 3

16. Why should we not taste unknown chemicals to test whether they are acids or bases? 3

17. Why does a tooth decay get worse with acid? 3

OR

Make a table of the colours of three indicators with acid and base. 3

18. Explain how red cabbage extract can be used as an indicator. 3

OR

Why is rinsing the mouth after eating sweets helpful? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to classify given household substances as acidic, basic or neutral using litmus, turmeric and China rose. 5

OR

Explain acids found in food items and the natural source of each. 5

20. Explain neutralisation with the example of an acid and a base. Write the equation and state two applications. 5

OR

Explain how a student can test whether a soil sample is acidic, basic or neutral. 5

END OF QUESTION PAPER — Total: 40 Marks

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